What is the full form of computer ?

Common Operating Machine Particularly used for Technical and Educational Research.

Why do computers only use 0's and 1's?

Computers uses transistors that takes either high (1) or low (0) voltage
Transistors combine together to form logical gates(AND, OR,NOT).

Logical gates combine together to form digital circuits (ADDER, MULTIPLEXER).

These circuits combine together to form ALU.

The logical gates also form FLIP-FLOP, these FLIP-FLOP combine together to form registers which is a memory unit.

Flip-Flop usually stores 1 bit of data, and register stores 1 byte of data (8 bits).

These registers and circuits combine and work together inside the ALU.

Why JavaScript is the most popular language ?

JavaScript is the most popular programming language primarily because of its **ubiquity as the only language supported by all web browsers**, its **versatility for both front-end and back-end development**, and an extensive, supportive community and ecosystem of tools.

JavaScript is introduced in December 1995 for the Netscape browser.

What can in-browser JavaScript do ?

Interact with HTML to add new content in the web page.

React to user actions like mouse click

Sends request over the network to remote server.

Uses local storage to store some data.

Get and set cookies.

Most JS Developers uses two main type of editors,

- IDE
- Lightweight Editors

IDE -> Visual studio code, webstorm

Visual studio code -> Free

Visual studio -> Paid

Variables in JavaScript

In JavaScript, variables can be used to store data.

A variable name can contain letters, numbers and special characters like $ and _.

The variable name should not start with number.

Can’t use a reserved name like return or interface to be as a variable name.

Use upper case variable name for constant variables.

E.g : const DOB = “18.09.2005”.

Data Types in JavaScript

There are 8 different data types in javaScript, but variable are not bound to them. i.e, we can store any data type like string or number in a data type.

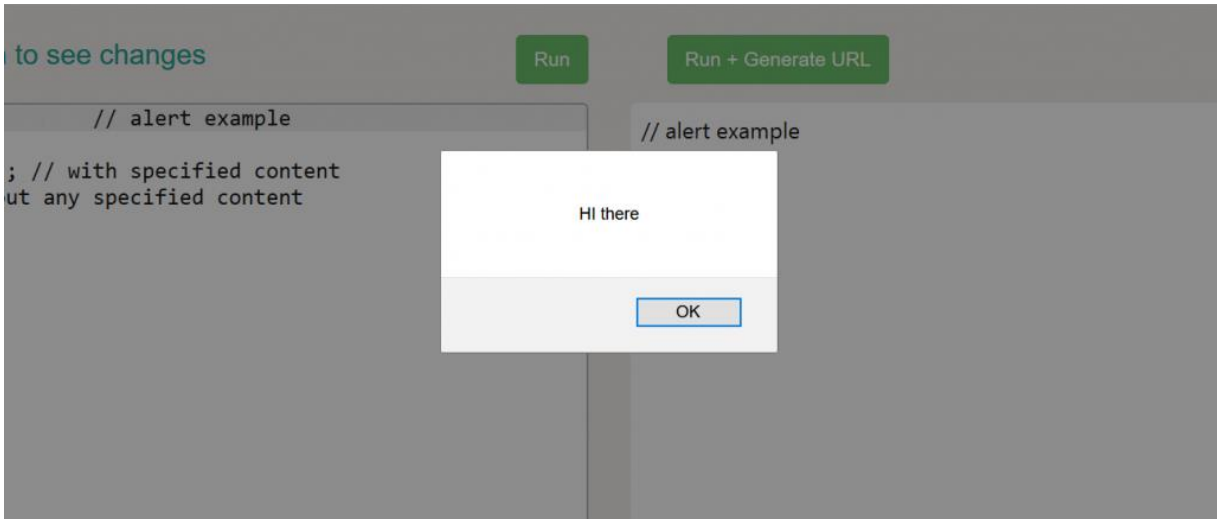
7 primitive data types ->

1. Number
2. BigInt
3. Boolean
4. String
5. Null
6. Undefined
7. Symbol

1 Non- primitive data type -> Object

Interaction: alert, prompt, confirm

Alert -> gives a modal window, where the alert message is displayed.



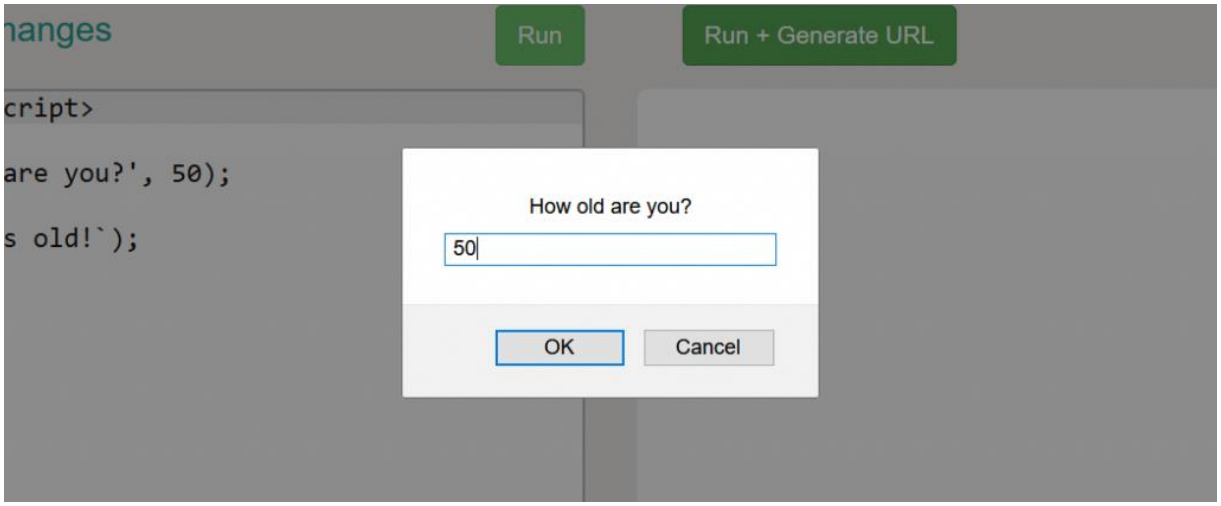
Prompt -> It shows a modal with a text to show the visitor, input field for the visitor.

Eg:

// prompt example

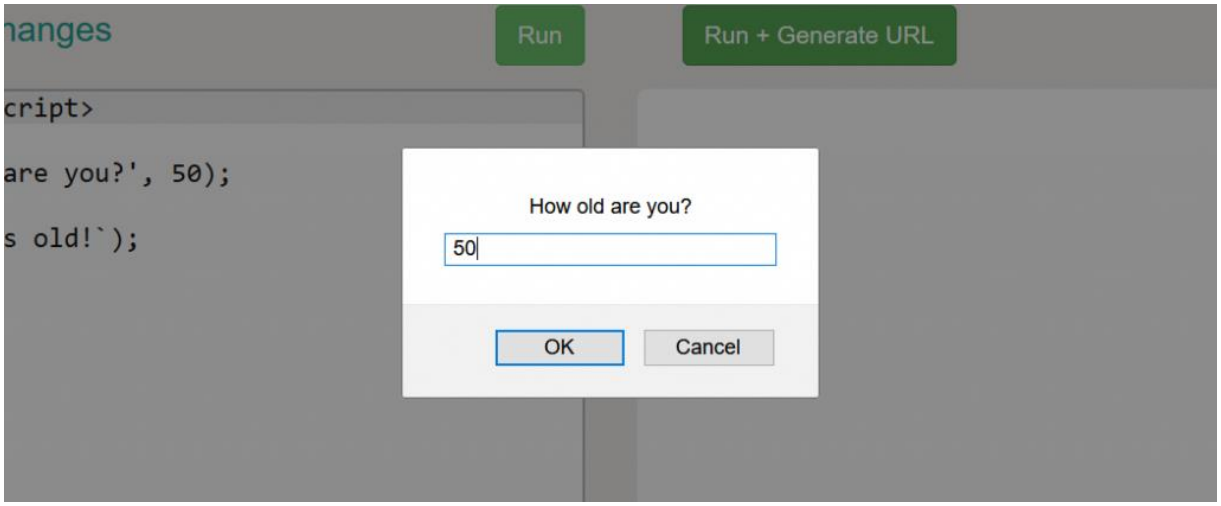
let age = prompt('How old are you?', 50);

alert(`You are ${age} years old!`);



If the user did not enter any input, then it returns null as output.

Confirm -> It shows a modal window with a question and two buttons.



Type Conversion:

String Conversion

Alert() automatically converts anything given into a string and we can also use String() function to convert into string.

Number Conversion

When applying division to non-numeric values, it automatically converts the result to number.

We can use Number() function to convert the output to number.

When the Number() function is applied to non-numeric value it will result in Nan.

Number(null) -> 0

Boolean Conversion

Boolean() function is used to convert into Boolean.

Values that are intuitively empty like 0, Nan, “ ”,null, undefined are false.

Other values are true.

Sometimes a !!(value) can be used to convert a value into Boolean.